Name
Date
Course

Simplifying Algebraic Fractions

Pre-Algebra Review

① $\frac{10x^6}{15x^4} = \boxed{\frac{2x^2}{3}}$

② $\frac{-8y^2}{2y^5} = \boxed{\frac{-4}{y^3}}$

③ $\frac{22a^2b^6}{-33ab^{11}} = \boxed{\frac{2a}{-3b^5}}$

④ $\frac{-16x^9y}{-32x^6y^7} = \boxed{\frac{x^3}{2y^6}}$

⑤ $\frac{36x^5}{-9x^3} = \boxed{-4x^2}$

⑥ $\frac{-18y}{20y^5} = \boxed{\frac{-9}{10y^4}}$

⑦ $\frac{42ab}{63a^5b} = \frac{6}{9a^4} = \boxed{\frac{2}{3a^4}}$

⑧ $\frac{-2x^{10}yz^8}{-10xy^5z^9} = \boxed{\frac{x^9}{5y^4z}}$

Beginning Algebra

⑨ $\frac{x^2-3x-10}{x-5} = \frac{(x-5)(x+2)}{x-5} = (x+2) = \boxed{x+2}$

⑩ $\frac{x+6}{x^2+11x+30} = \frac{x+6}{(x+5)(x+6)} = \boxed{\frac{1}{x+5}}$

⑪ $\frac{x^2+9x+20}{x^2+5x+4} = \frac{(x+4)(x+5)}{(x+1)(x+4)} = \boxed{\frac{x+5}{x+1}}$

⑫ $\frac{x^2+4x-32}{x^2+x-20} = \frac{(x+8)(x-4)}{(x+5)(x-4)} = \boxed{\frac{x+8}{x+5}}$

⑬ $\frac{x+8}{x^2+13x+40} = \frac{x+8}{(x+5)(x+8)} = \boxed{\frac{1}{x+5}}$

⑭ $\frac{x^2-4x-12}{x-6} = \frac{(x-6)(x+2)}{x-6} = \boxed{x+2}$

⑮ $\frac{x^2+13x+36}{x^2+7x+12} = \frac{(x+4)(x+9)}{(x+4)(x+3)} = \boxed{\frac{x+9}{x+3}}$

⑯ $\frac{x^2+6x-16}{x^2+3x-10} = \frac{(x+8)(x-2)}{(x+5)(x-2)} = \boxed{\frac{x+8}{x+5}}$

⑰ $\frac{x^2-3x-10}{2x^2+7x+6} = \frac{(x-5)(x+2)}{(2x+3)(x+2)} = \boxed{\frac{x-5}{2x+3}}$

⑱ $\frac{x^2+x-30}{2x^2+15x+18} = \frac{(x+6)(x-5)}{(2x+3)(x+6)} = \boxed{\frac{x-5}{2x+3}}$

⑲ $\frac{x^2-9x+20}{3x^2-11x-20} = \frac{(x-5)(x-4)}{(3x+4)(x-5)} = \boxed{\frac{x-4}{3x+4}}$

⑳ $\frac{3x^2-10x-8}{x^2-5x+4} = \frac{(3x+2)(x-4)}{(x-4)(x-1)} = \boxed{\frac{3x+2}{x-1}}$

㉑ $\frac{x^2-25}{x^2+2x-35} = \frac{(x-5)(x+5)}{(x+7)(x-5)} = \boxed{\frac{x+5}{x+7}}$

(22) $\dfrac{x^2-49}{x^2+10x+21} = \dfrac{(x-7)(x+7)}{(x+7)(x+3)} = \boxed{\dfrac{x-7}{x+3}}$

(23) $\dfrac{4x^2-23x-35}{16x^2-25} = \dfrac{(4x+5)(x-7)}{(4x-5)(4x+5)} = \boxed{\dfrac{x-7}{4x-5}}$

(24) $\dfrac{9x^2-4}{3x^2+4x-4} = \dfrac{(3x-2)(3x+2)}{(3x-2)(x+2)} = \boxed{\dfrac{3x+2}{x+2}}$

(25) $\dfrac{x^2-11x+28}{x-4} = \dfrac{(x-7)(x-4)}{x-4} = \boxed{x-7}$

(26) $\dfrac{x+2}{x^2+5x+6} = \dfrac{x+2}{(x+2)(x+3)} = \boxed{\dfrac{1}{x+3}}$

(27) $\dfrac{x^2-10x+21}{x-3} = \dfrac{(x-7)(x-3)}{x-3} = \boxed{x-7}$

(28) $\dfrac{x+2}{x^2+6x+8} = \dfrac{x+2}{(x+2)(x+4)} = \boxed{\dfrac{1}{x+4}}$

The Negative One Rule

(29) $\dfrac{x^2-81}{9-x} = \dfrac{(x-9)(x+9)}{9-x} = \dfrac{(x-9)(x+9)}{-1(-9+x)} = \dfrac{x+9}{-1}$

$= -\dfrac{(x+9)}{1}$

(30) $\dfrac{x^2-8x+16}{4-x} = \dfrac{(x-4)(x-4)}{4-x} = \dfrac{(x-4)(x-4)}{-1(-4+x)}$

$= \dfrac{x-4}{-1} = \boxed{-x+4}$

(31) $\dfrac{x^2-64}{8-x} = \dfrac{(x+8)(x-8)}{-1(-8+x)} = \dfrac{x+8}{-1} = \boxed{-x-8}$

(32) $\dfrac{3-x}{x^2-4x+3} = \dfrac{-(-3+x)}{(x-3)(x-1)} = \boxed{\dfrac{-1}{x-1}}$

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(33) $\dfrac{4x^7+44x^6+112x^5}{2x^5+2x^4-24x^3} = \dfrac{4x^5(x^2+11x+28)}{2x^3(x^2+x-12)}$

$= \dfrac{4x^5(x+7)(x+4)}{2x^3(x+4)(x-3)}$

$= \boxed{\dfrac{2x^2(x+7)}{x-3}}$

(34) $\dfrac{6x^4-6x^3-36x^2}{-4x^7+12x^6} = \dfrac{6x^2(x^2-x-6)}{-4x^6(x-3)}$

$= \dfrac{6x^2(x-3)(x+2)}{-4x^6(x-3)}$

$= \boxed{-\dfrac{3(x+2)}{2x^4}}$

(35) $$\frac{6x^7+36x^6+48x^5}{2x^4-2x^3-12x^2} = \frac{6x^5(x^2+6x+8)}{2x^2(x^2-x-6)} = \frac{6x^5(x+4)(x+2)}{2x^2(x-3)(x+2)} = \boxed{\frac{3x^3(x+4)}{x-3}}$$

(36) $$\frac{12x^3-48x}{16x^4-24x^3-16x^2} = \frac{12x(x^2-4)}{8x^2(2x^2-3x-2)} = \frac{12x(x-2)(x+2)}{8x^2(2x+1)(x-2)} = \boxed{\frac{3(x+2)}{2x(2x+1)}}$$

(37) $$\frac{10x^9+10x^8-120x^7}{-8x^4+24x^3} = \frac{10x^7(x^2+x-12)}{-8x^3(x-3)} = \frac{10x^7(x+4)(x-3)}{-8x^3(x-3)} = \boxed{-\frac{5x^4(x+4)}{4}}$$

(38) $$\frac{18x^4-27x^3-18x^2}{12x^6-48x^4} = \frac{9x^2(2x^2-3x-2)}{12x^4(x^2-4)} = \frac{9x^2(2x+1)(x-2)}{12x^4(x-2)(x+2)} = \boxed{\frac{3(2x+1)}{4x^2(x+2)}}$$

(39) $$\frac{-4x^5-4x^4+8x^3}{8x^3-24x^2+16x} = \frac{-4x^3(x^2+x-2)}{8x(x^2-3x+2)} = \frac{-4x^3(x+2)(x-1)}{8x(x-2)(x-1)} = \boxed{-\frac{x^2(x+2)}{2(x-2)}}$$

(40) $$\frac{16x-4x^2}{-x^{11}+12x^{10}-32x^9} = \frac{-4x(-4+x)}{-x^9(x^2-12x+32)} = \frac{-4x(-4+x)}{-x^9(x-8)(x-4)} = \boxed{\frac{4}{x^8(x-8)}}$$

(41) $$\frac{-2x^5+6x^4-4x^3}{8x^3-48x^2+64x} = \frac{-2x^3(x^2-3x+2)}{8x(x^2-6x+8)} = \frac{-2x^3(x-2)(x-1)}{8x(x-2)(x-4)} = \boxed{-\frac{x^2(x-1)}{4(x-4)}}$$

(42) $$\frac{8x-2x^2}{-x^{11}+12x^{10}-32x^9} = \frac{-2x(-4+x)}{-x^9(x^2-12x+32)} = \frac{-2x(-4+x)}{-x^9(x-4)(x-8)} = \boxed{\frac{2}{x^8(x-8)}}$$

(43) $\dfrac{x^4+13x^2+40}{2x^5+10x^3} = \dfrac{(x^2+5)(x^2+8)}{2x^3(x^2+5)}$

$= \boxed{\dfrac{x^2+8}{2x^3}}$

(44) $\dfrac{3x^6+6x^4}{x^4+11x^2+18} = \dfrac{3x^4(x^2+2)}{(x^2+9)(x^2+2)}$

$= \boxed{\dfrac{3x^4}{x^2+9}}$

(45) $\dfrac{5x^4-2x^2}{5x^6-22x^4+8x^2} = \dfrac{x^2(5x^2-2)}{x^2(5x^4-22x^2+8)}$

$= \dfrac{x^2(5x^2-2)}{x^2(5x^2-2)(x^2-4)}$

$= \dfrac{1}{x^2-4}$

$= \boxed{\dfrac{1}{(x-2)(x+2)}}$

(46) $\dfrac{x^3+8x}{2x^5+10x^3-48x} = \dfrac{x(x^2+8)}{2x(x^4+5x^2-24)}$

$= \dfrac{x(x^2+8)}{2x(x^2+8)(x^2-3)}$

$= \boxed{\dfrac{1}{2(x^2-3)}}$

(47) $\dfrac{x^2+10xy+16y^2}{3x^3y+24x^2y^2} = \dfrac{(x+2y)(x+8y)}{3x^2y(x+8y)}$

$= \boxed{\dfrac{x+2y}{3x^2y}}$

(48) $\dfrac{a^2+9ab+14b^2}{3a^3b+21a^2b^2} = \dfrac{(a+2b)(a+7b)}{3a^2b(a+7b)}$

$= \boxed{\dfrac{a+2b}{3a^2b}}$

(49) $$\frac{20x^{11}-85x^9+20x^7}{-12x^7+14x^6+10x^5} = \frac{5x^7(4x^4-17x^2+4)}{-2x^5(6x^2-7x-5)}$$

$$= \frac{5x^7(x^2-4)(4x^2-1)}{-2x^5(3x-5)(2x+1)}$$

$$= \frac{5x^7(x-2)(x+2)(2x-1)(2x+1)}{-2x^5(3x-5)(2x+1)}$$

$$= \boxed{\frac{5x^2(x-2)(x+2)(2x-1)}{-2\ (3x-5)}}$$

(50) $$\frac{12x^3+42x^2+36x}{16x^{11}-100x^9+144x^7} = \frac{6x(2x^2+7x+6)}{4x^7(4x^4-25x^2+36)}$$

$$= \frac{6x(2x+3)(x+2)}{4x^7(4x^2-9)(x^2-4)}$$

$$= \frac{6x(2x+3)(x+2)}{4x^7(2x-3)(2x+3)(x-2)(x+2)}$$

$$= \boxed{\frac{3}{2x^6(2x-3)(x-2)}}$$

(51) $$\frac{3x^3+2x^2+12x+8}{3x^2-25x-18} = \frac{x^2(3x+2)+4(3x+2)}{(3x+2)(x-9)}$$

$$= \frac{(3x+2)(x^2+4)}{(3x+2)(x-9)}$$

$$= \boxed{\frac{x^2+4}{x-9}}$$

(52) $$\frac{3x^2-19x-14}{3x^3+2x^2+9x+6} = \frac{(3x+2)(x-7)}{x^2(3x+2)+3(3x+2)}$$

$$= \frac{(3x+2)(x-7)}{(3x+2)(x^2+3)}$$

$$\boxed{\frac{x-7}{x^2+3}}$$

(53) $$\frac{3-x}{2x^3-6x^2+5x-15} = \frac{-1(-3+x)}{2x^2(x-3)+5(x-3)}$$

$$= \frac{-1(-3+x)}{(x-3)(2x^2+5)}$$

$$= \boxed{\frac{-1}{2x^2+5}}$$

(54) $\dfrac{2x^3-4x^2+3x-6}{2-x} = \dfrac{2x^2(x-2)+3(x-2)}{-(-2+x)}$

$= \dfrac{(x-2)(2x^2+3)}{-(-2+x)}$

$= \dfrac{2x^2+3}{-1}$

$= -1(2x^2+3)$

$= \boxed{-2x^2-3}$